

ArchGen AI

Cloud-Native AI Architecture Generation Platform on Microsoft Azure

- ✓ Cloud-Native Azure Deployment
- ✓ Infrastructure as Code using Terraform
- ✓ Azure App Services & Cosmos DB
- ✓ Ollama LLM on Azure VM
- ✓ Secure Networking & Application Gateway
- ✓ Enterprise-Grade WAF Protection

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| Cloud Implementation Objectives



SaaS Scalability

Build a scalable cloud-native platform using Azure's managed services for elastic growth.



IaC Automation

Eliminate manual configuration by automating the entire resource lifecycle via Terraform.



Zero Trust Network

Implement secure segmentation using VNets, Private Endpoints, and WAF protection.



AI Workloads

Host and manage Ollama LLM effectively on optimized Azure Virtual Machines.



Design Consistency

Improve deployment speed and consistency with reusable architecture modules.



Business Goal

Provide an AI-driven platform for automated cloud architecture recommendations.

Azure Cloud Architecture

Architecture Components

- ✓ **Application Gateway:** Traffic routing with WAF protection.
- ✓ **Frontend & Backend:** Hosted on Azure App Services.
- ✓ **Database:** Azure Cosmos DB (Mongo API) for NoSQL data.
- ✓ **AI Engine:** Ollama LLM running on specialized Azure VM.
- ✓ **Connectivity:** Private Endpoints & VNet Peering for isolation.
- ✓ **Management:** Azure Bastion & NAT Gateway for secure access.

Designed for high availability and isolation, the platform leverages Private Link to keep all traffic off the public internet.

 Azure Architecture Diagram

Azure Services Utilized

Service Category	Azure Service	Strategic Purpose
Compute	App Service / VM	Application hosting and AI model execution
Network	VNet / Bastion	Isolation and secure administrative entry
Security	App Gateway / WAF	L7 load balancing and threat protection
Database	Cosmos DB	Scalable NoSQL storage with Mongo API
Connectivity	Private Endpoint	Secure internal service communication
Management	Resource Group	Logical grouping and lifecycle management



Leveraging first-party Azure services to ensure seamless integration and SLA compliance.

Infrastructure as Code with Terraform



Modular Project Structure

```
terraform/  
├── main.tf  
├── variables.tf  
├── outputs.tf  
└── modules/  
    ├── network/  
    ├── app_service/  
    ├── cosmosdb/  
    └── vm/
```

Code



Plan



Apply

Key Implementation Benefits

Automated Provisioning: Human-error reduction via declarative code.

Reusable Modules: Standardized "building blocks" for all environments.

Version Control: Infrastructure history tracked via Git for full auditability.

| Security and Network Architecture



Network Security

- VNet Segmentation
- Private Endpoints
- Private DNS Resolution



App Security

- HTTPS Enforcement
- JWT Auth Integration
- WAF Layer 7 Filtering



Infra Security

- Azure Bastion Access
- NAT Gateway Egress
- Restricted DB Firewalls

DevOps Centric Security



Security is integrated into the **Terraform** modules, ensuring that every deployment adheres to organizational compliance standards automatically. This "Shift-Left" approach guarantees secure defaults for every environment.

Project Outcome

- ✓ Designed Enterprise Azure Architecture
- ✓ Implemented Terraform-Based Infra
- ✓ Integrated AI Workloads (Ollama)
- ✓ Deployed Secure SaaS Platform
- ✓ Advanced Azure Networking
- ✓ Automated Cloud Provisioning

"ArchGen AI demonstrates the practical implementation of Cloud Architecture, IaC, Security Best Practices, and DevOps Principles on Microsoft Azure."

Thank You

Questions & Discussion

| Image Sources



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